The following example will illustrate our guidelines for modelling requirements of the problem domain with entity relationship diagrams. Using the summary paragraph of the problem description below, we will progress through the steps of design an ER model for the scenario

A company named XYZ has a number of data requirements as listed below. The company keeps track of its employee, departments and projects. After requirements were collected, the database analysts and designers came up with the following description of Mini world (part of the company to be represented in the database).

1. The company is organized into departments and for such, the company captures information such as Department Code, Department Name and the employee who manages the department. The department code uniquely identifies each instance of the department. A department may have several locations. The company also keeps track of the start date when the employee began managing the department. An employee can manage zero or many departments while a given department can be managed by one and only one employee
2. Information kept about employees includes Employee number, mobile, Gender, name, email, date of birth. Employee number uniquely identifies each instance of the department. An employee can have 1 or more mobile number and email address.
3. A department controls a number of projects. Information captured about projects include Project Code, Project Name and location. The project code uniquely identifies each instance of the project. A project can only have a single location.

4. A department can control one or more projects while a given project can be controlled by only one department. An employee can work on several projects which are not necessarily controlled by the same department while a given project can have one or more employees working on it. The company also keeps track of the number of hours per week that an employee works on each project. The company also keeps track of the direct supervisor of each employee.
5. Xyz keeps track of the defendants of each employee for insurance purpose. For this they need fist name of the dependant, his/her gender, birthdate and relationship to the employee.

Required:

Produce an ER model using

1. Crow feet Notation
2. Chen Notation